

Distributed Quantum Computing

Massimo Equi

Aalto University

15 June 2026

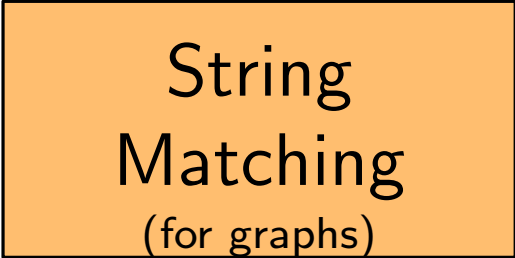
Interview for IQM

About Me

Three main research topics

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An orange rectangular box with a black border, containing the text "String Matching (for graphs)".

String
Matching
(for graphs)

About Me

Three main research topics

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Quantum
Computing

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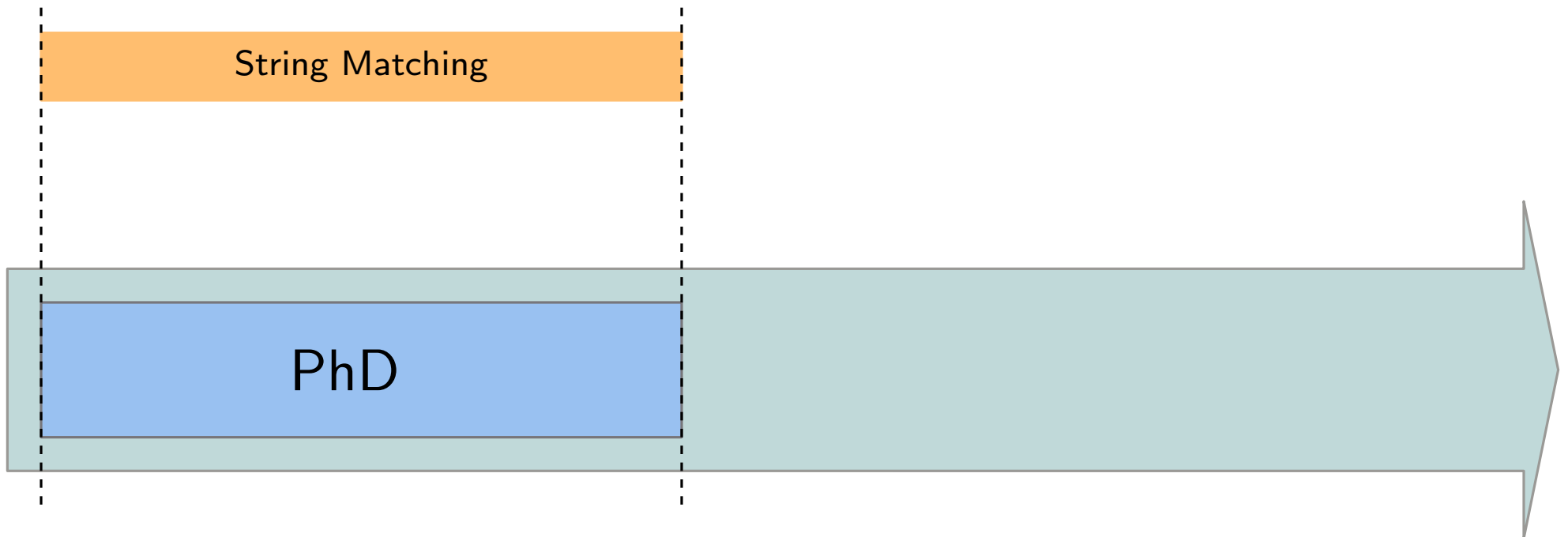
Distributed
Computing

String Matching

PhD

2019

2022



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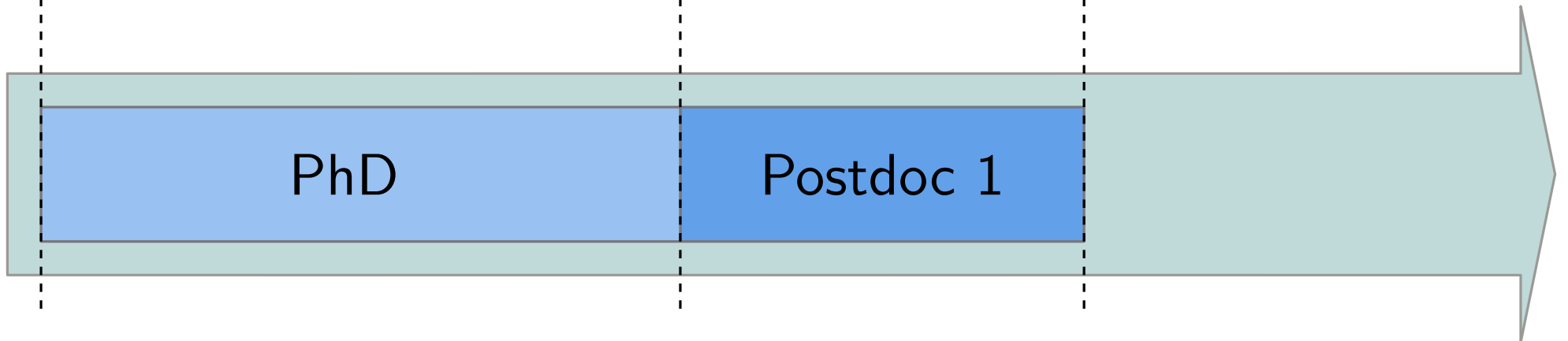
PhD

Postdoc 1

2019

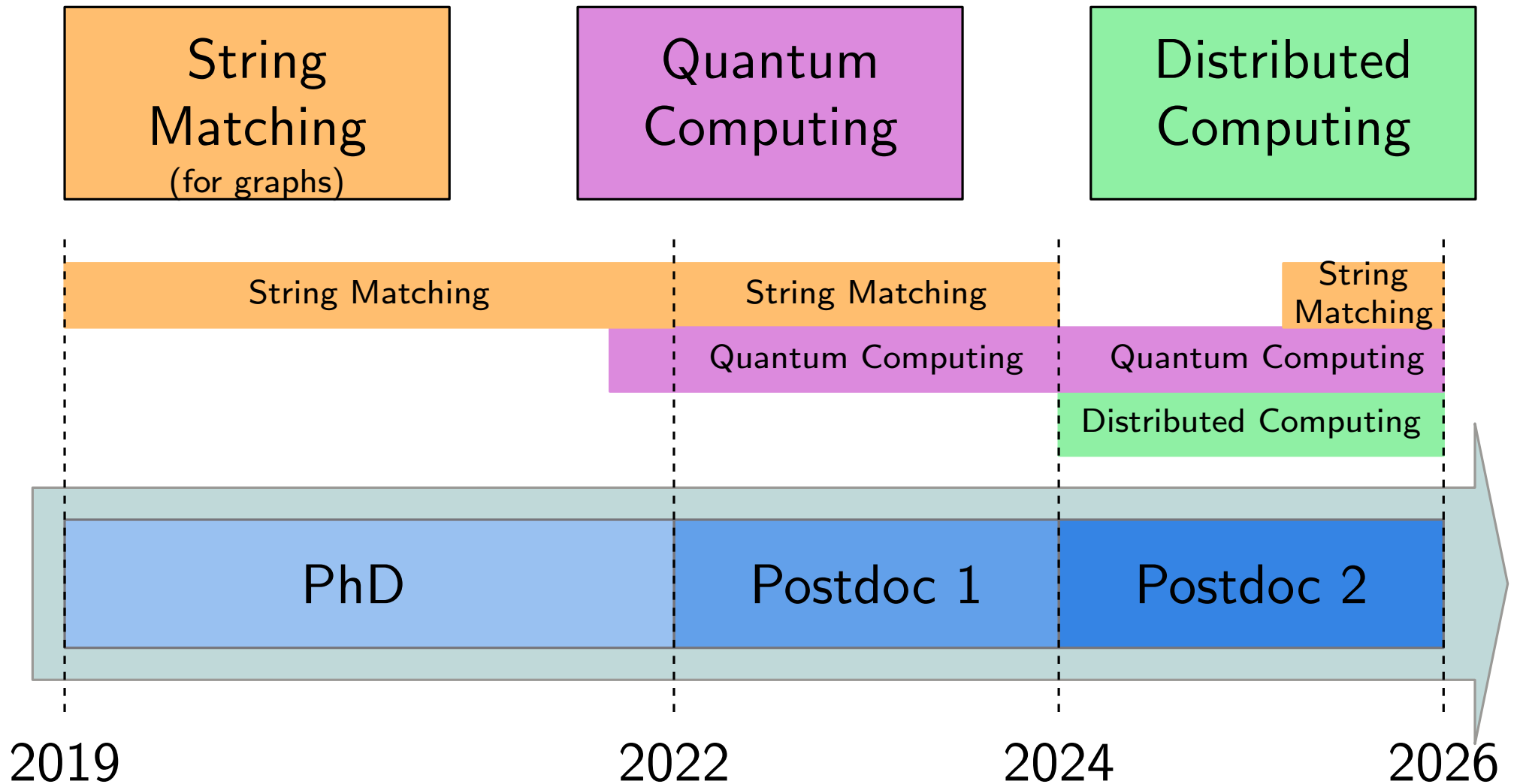
2022

2024

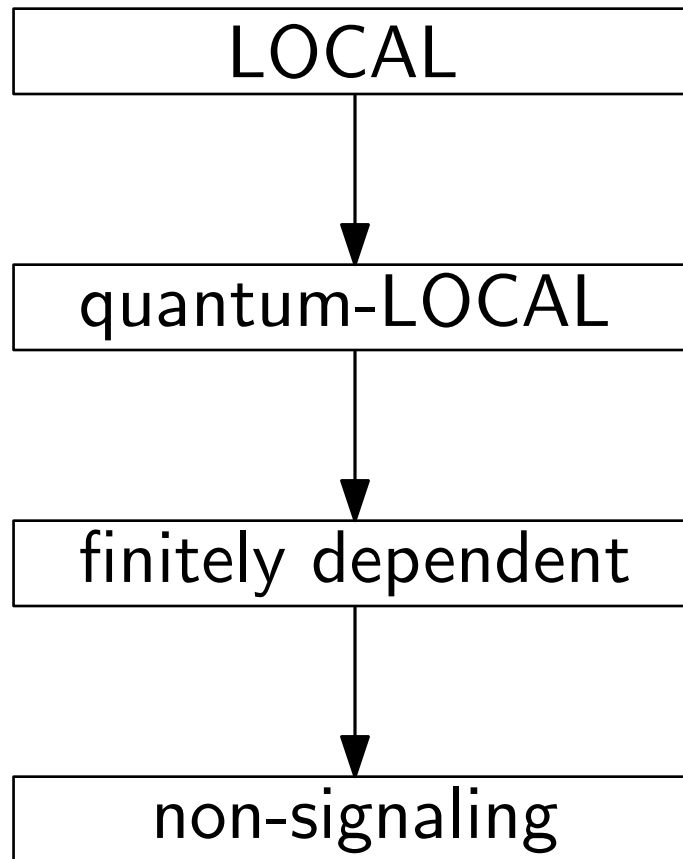


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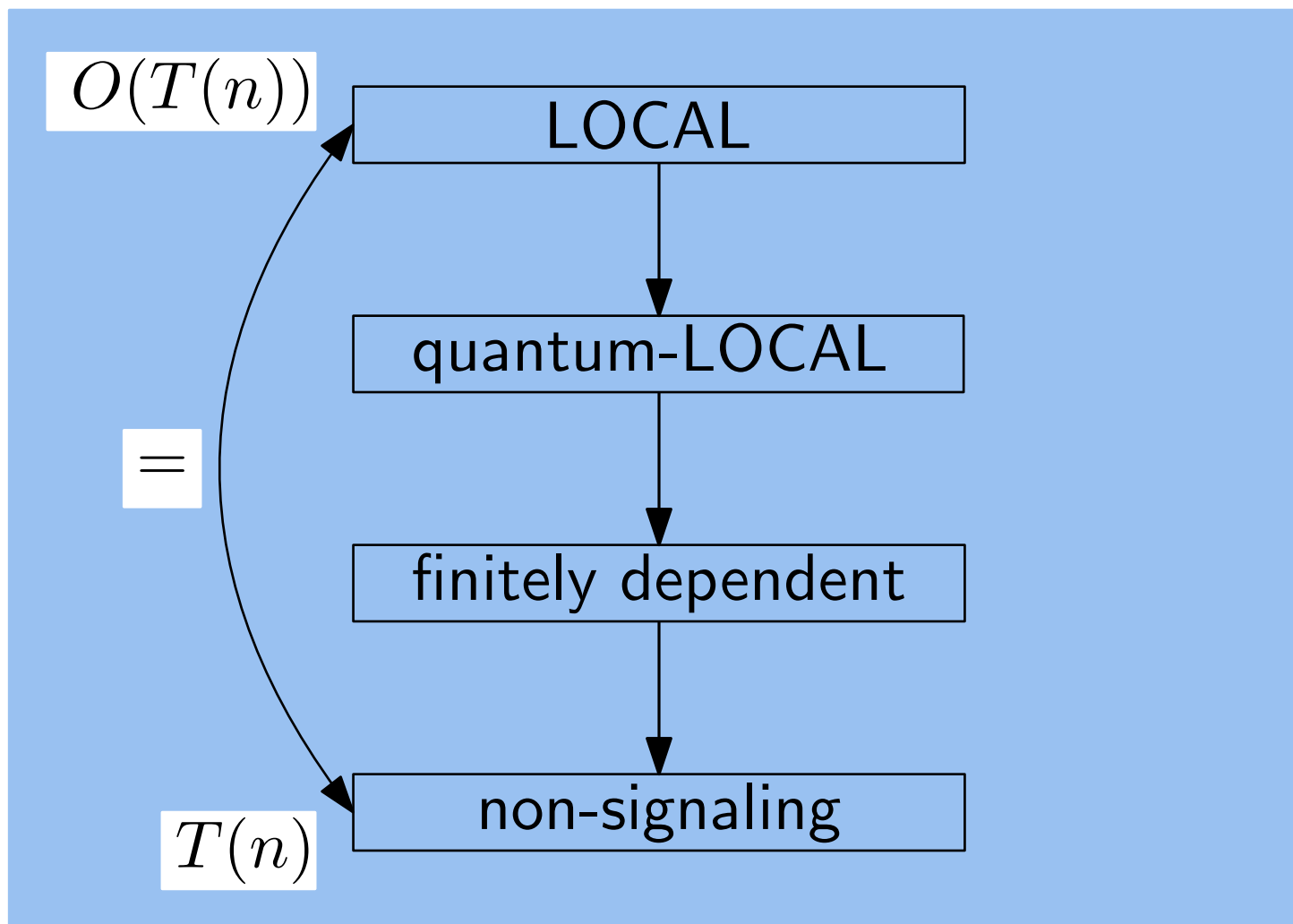


Distributed Hierarchy



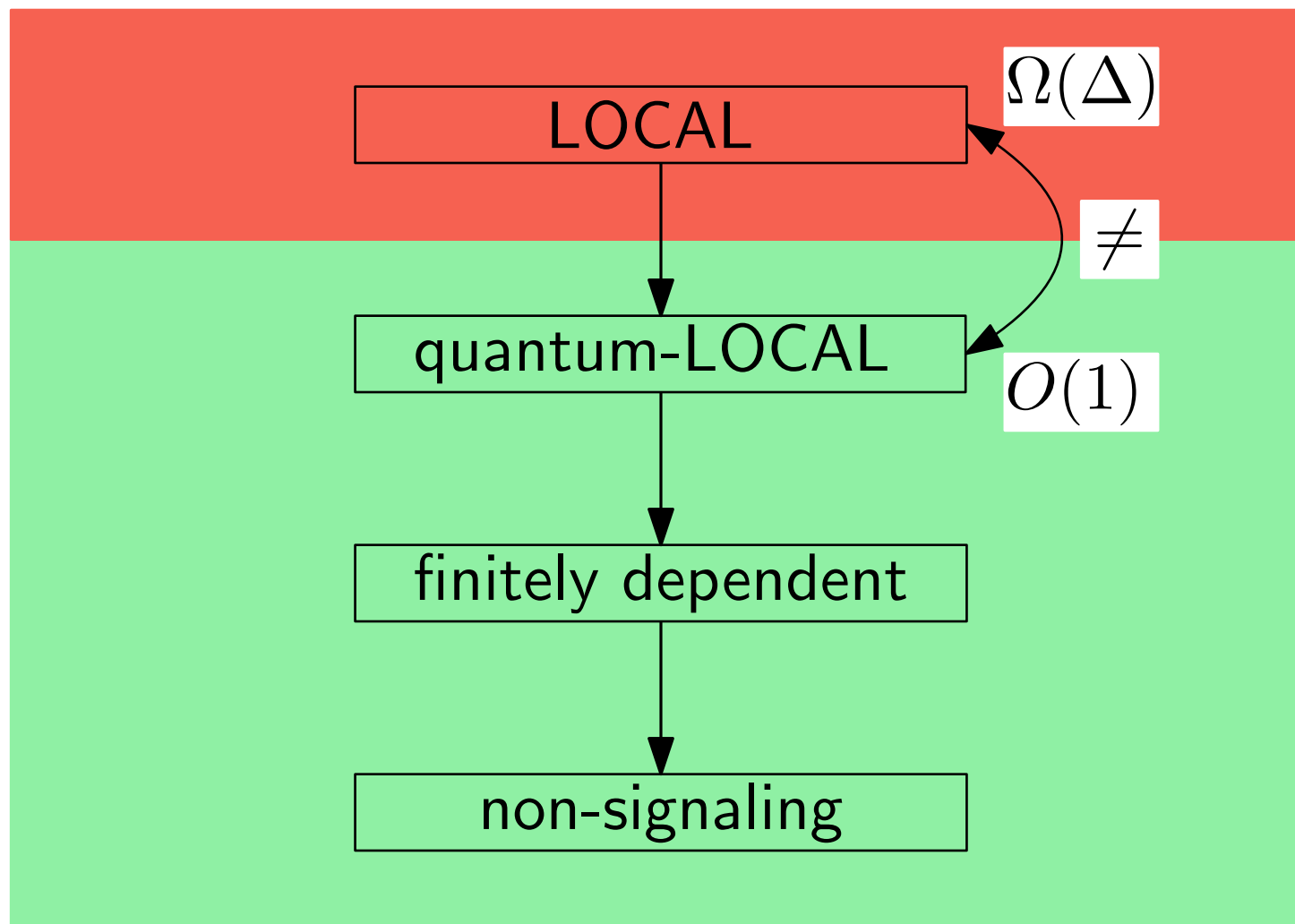
Distributed Hierarchy

For Linear Programs (LPs)

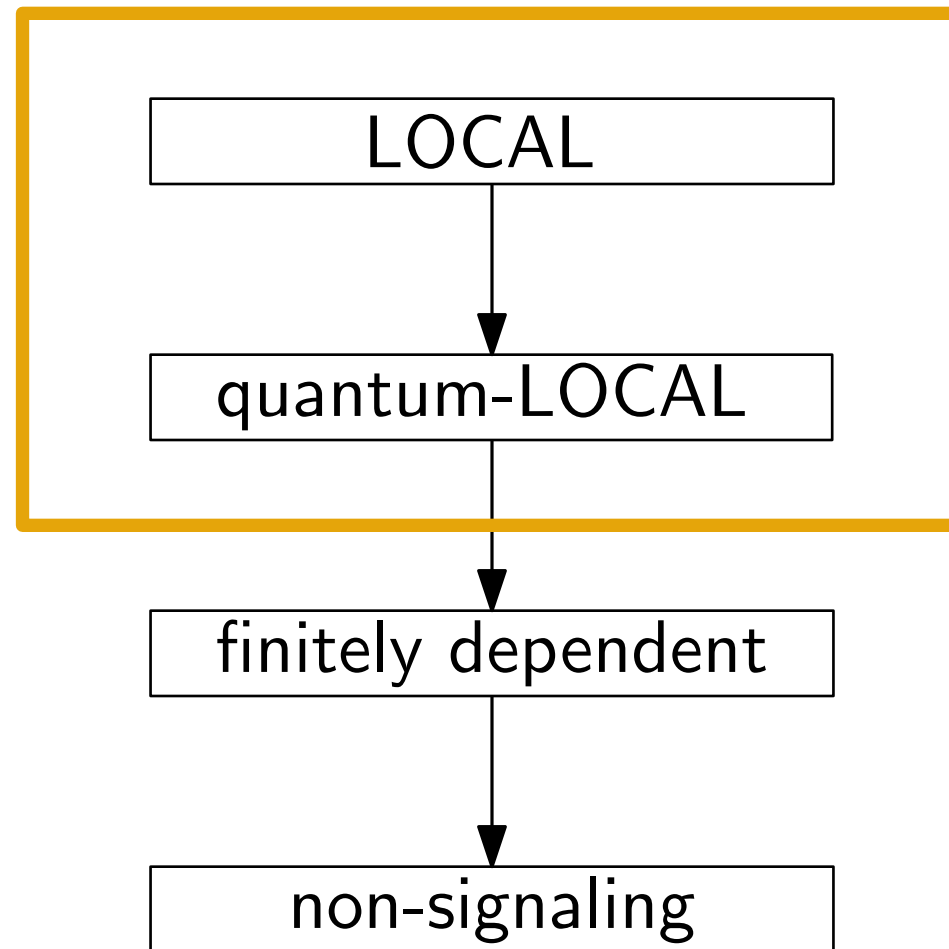


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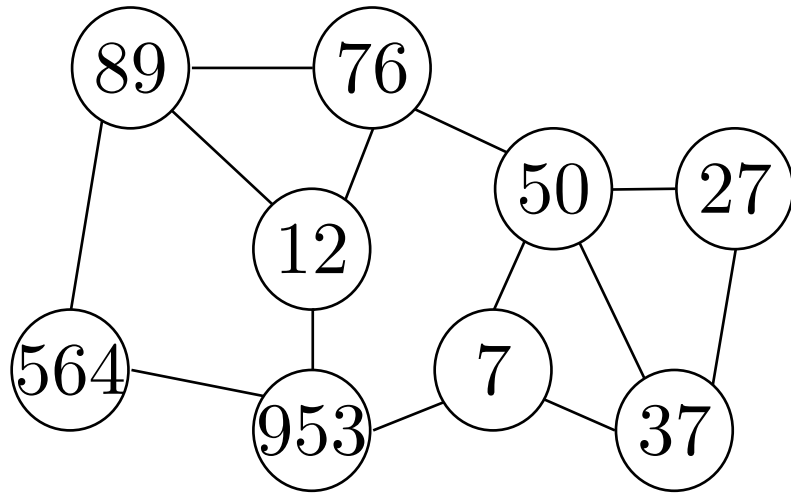
For Locally Checkable Labelling Problems (LCLs)



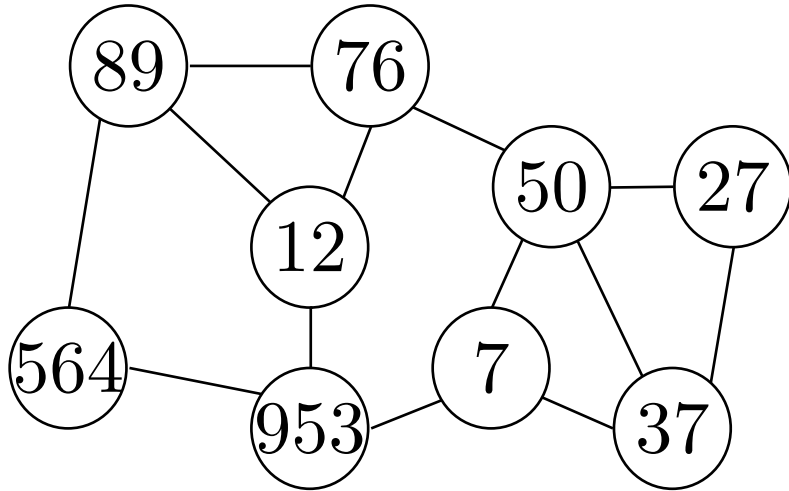
Distributed Hierarchy



LOCAL and quantum-LOCAL

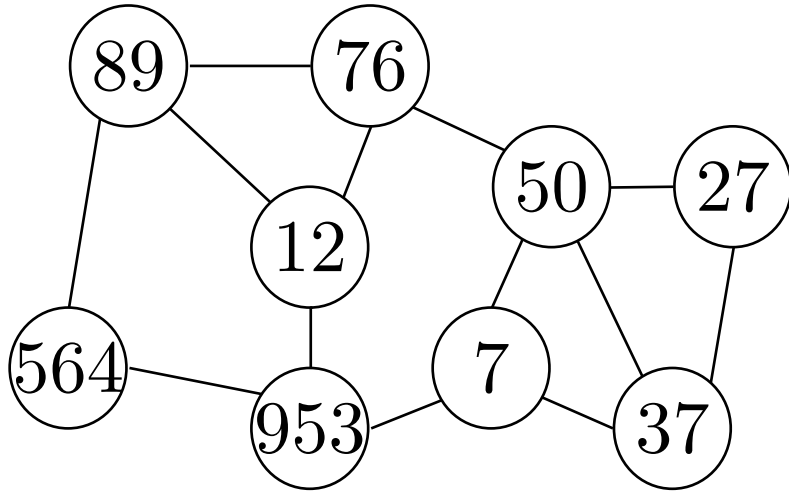


LOCAL and quantum-LOCAL



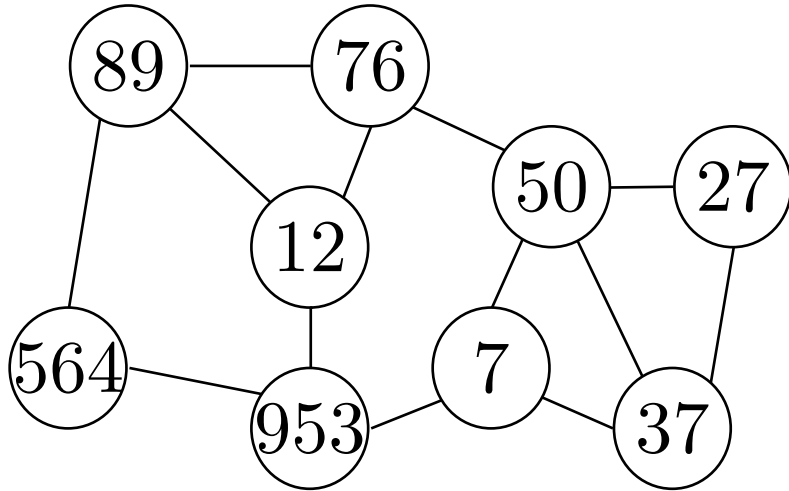
- The network is a graph $G = (V, E)$ of n nodes

LOCAL and quantum-LOCAL



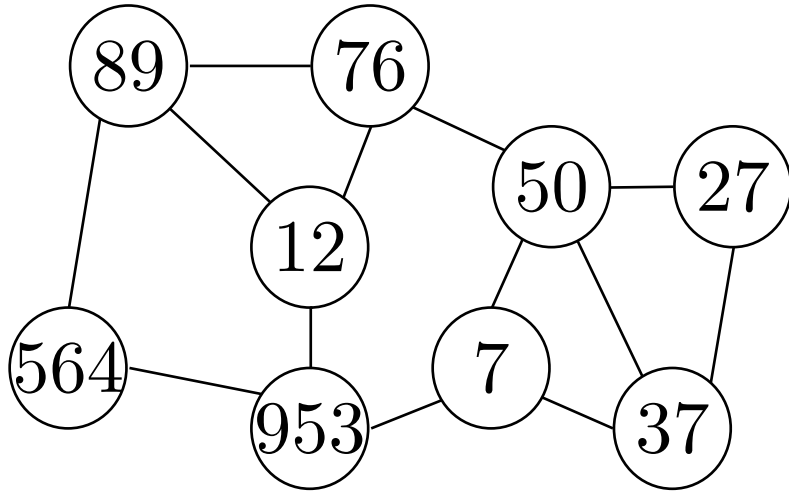
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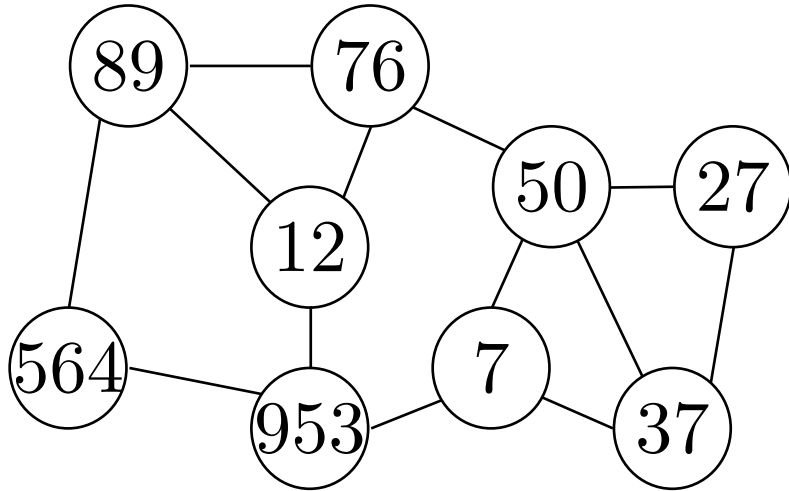
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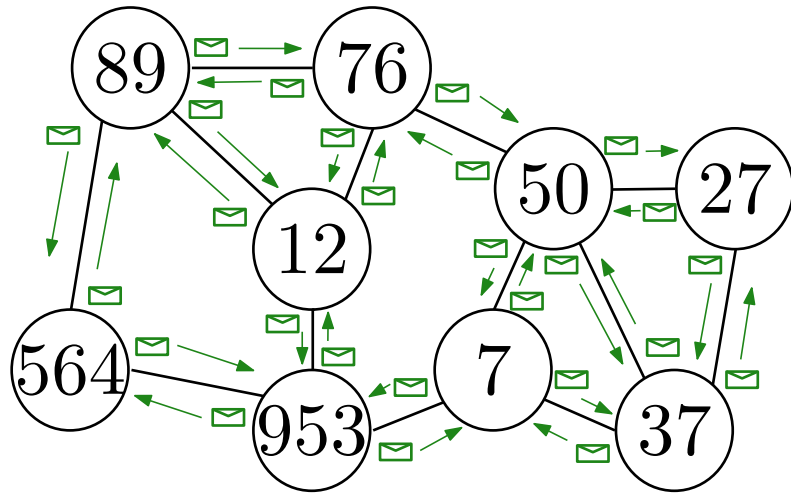
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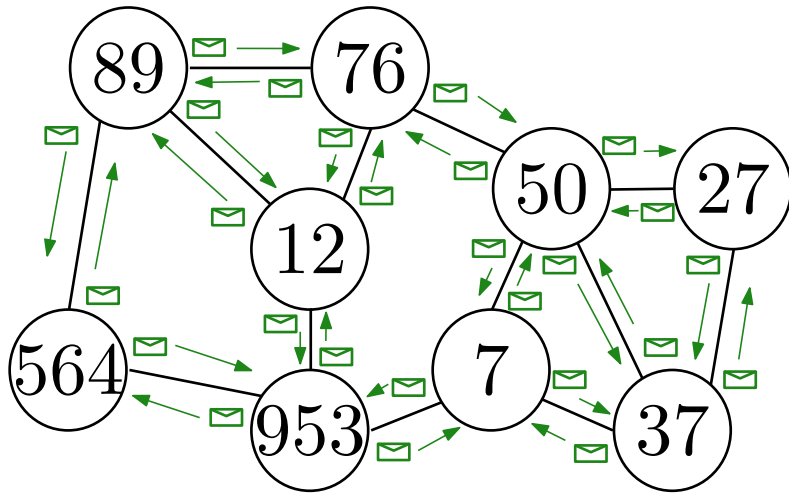


Complexity

Number of communication rounds as a function of n

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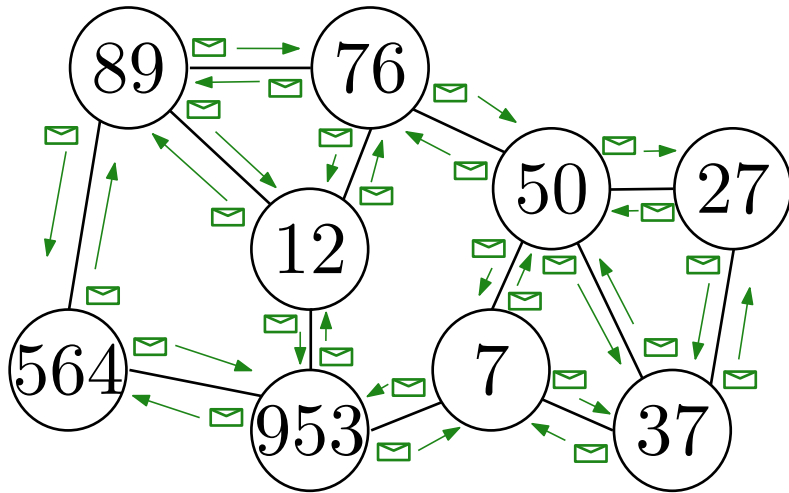
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Number of communication rounds as a function of n

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Nodes hold, send and receive **bits**

LOCAL and quantum-LOCAL



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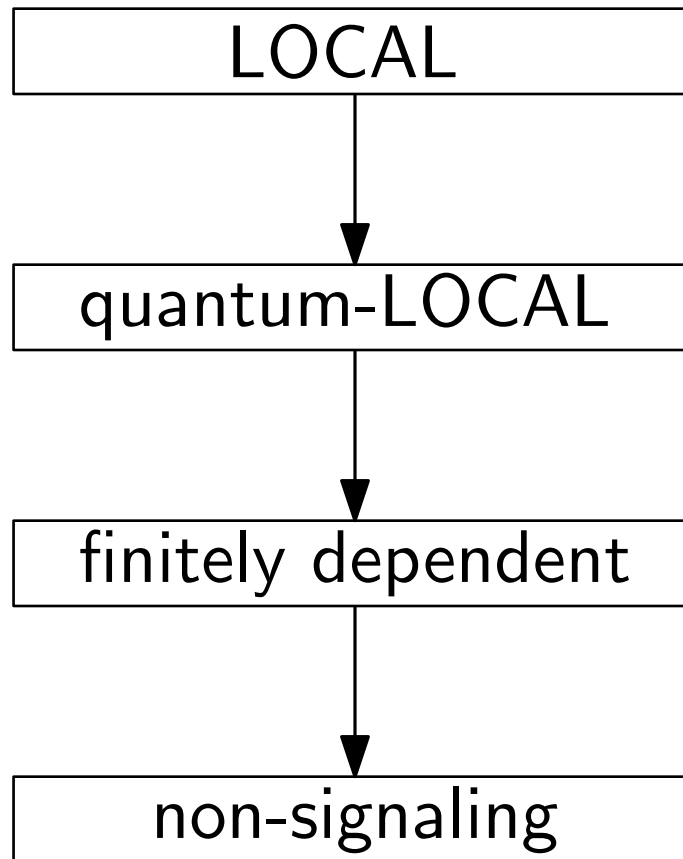
quantum-LOCAL

Nodes hold, send and receive **qubits**

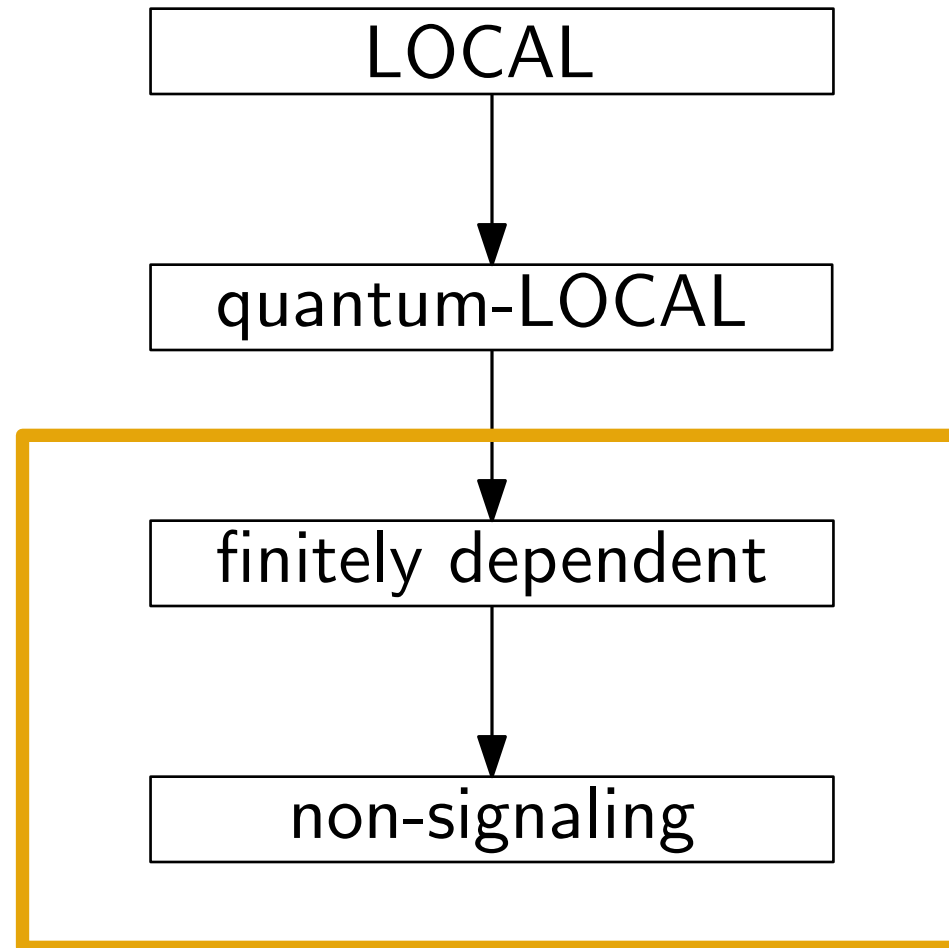
IDs are not needed

“send” qubits means applying SWAP gates

Distributed Hierarchy

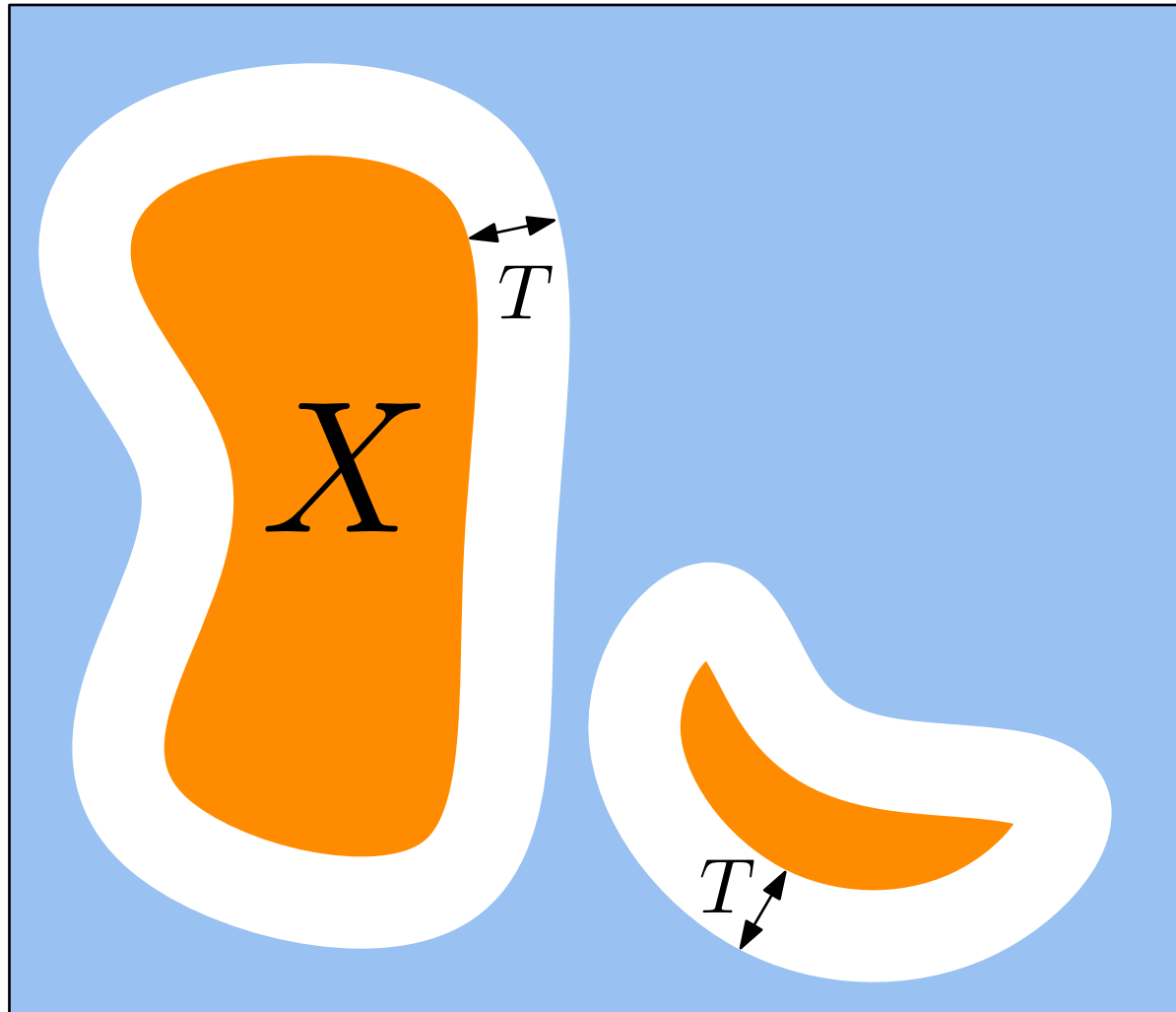


Distributed Hierarchy



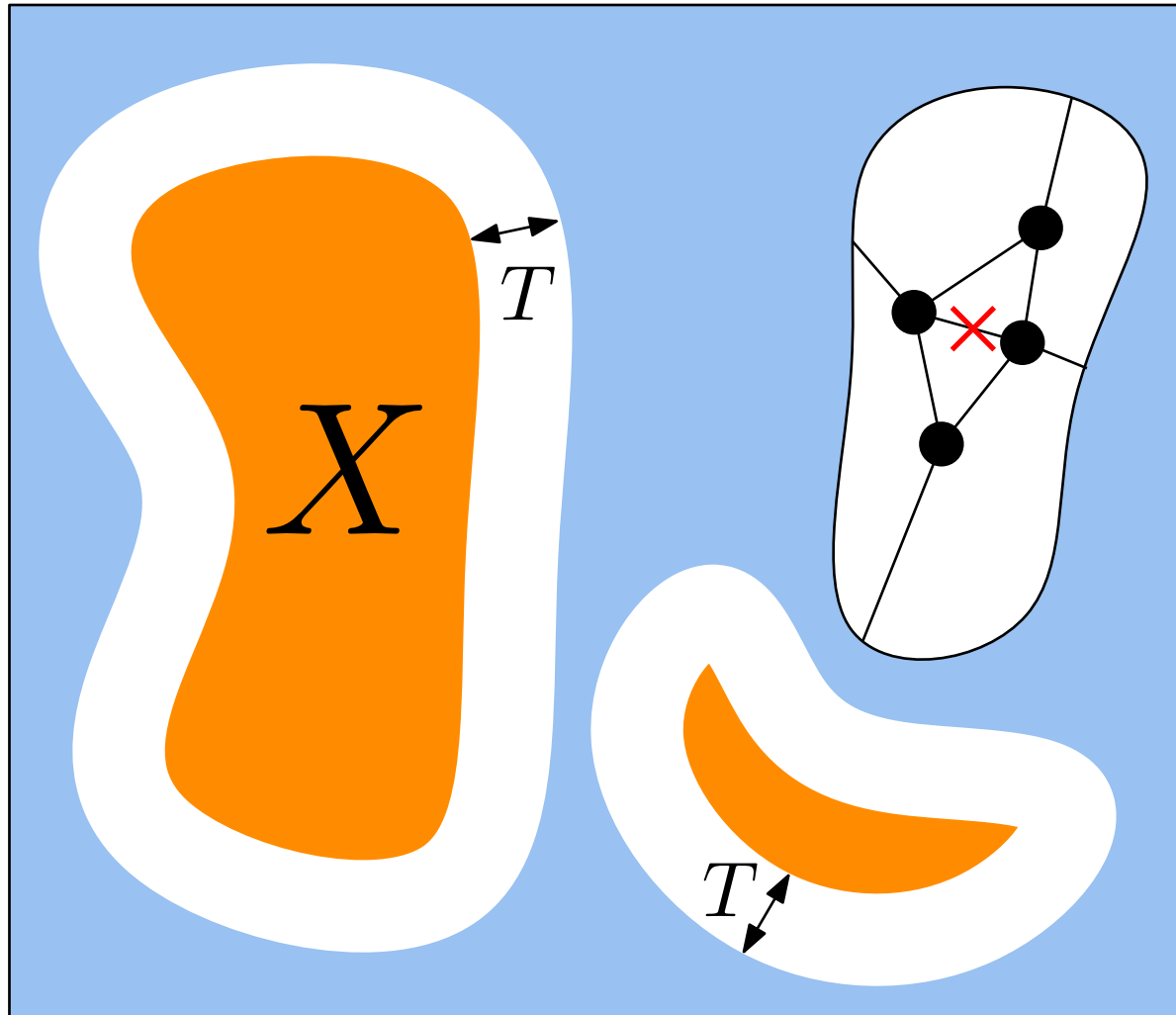
Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



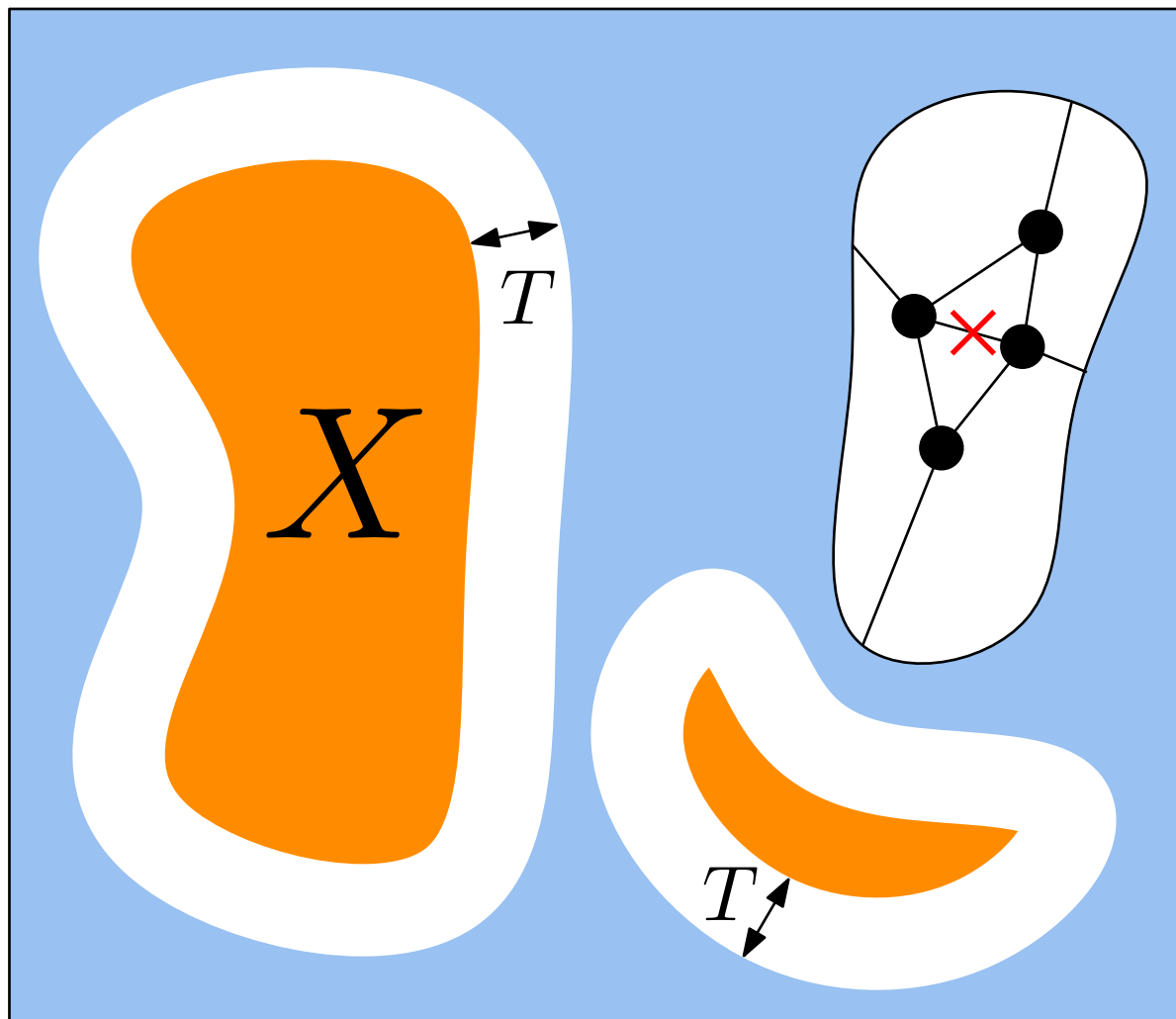
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Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

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Example of Distributed LP: **fractional maximum independent set**

$$\text{maximize } \sum_{v \in V} x_v$$

$$\text{subject to } \sum_{(v,u) \in E} x_v + x_u \leq 1 \quad \forall v \in V$$

$$0 \leq x_v \leq 1 \quad \forall v \in V$$

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$$0 \leq x_v \leq 1 \quad \forall v \in V$$

When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of maximum cardinality.

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Let $X = \sum_{v \in V} X_v$ be the rand. var. for the output of the entire graph

Key facts

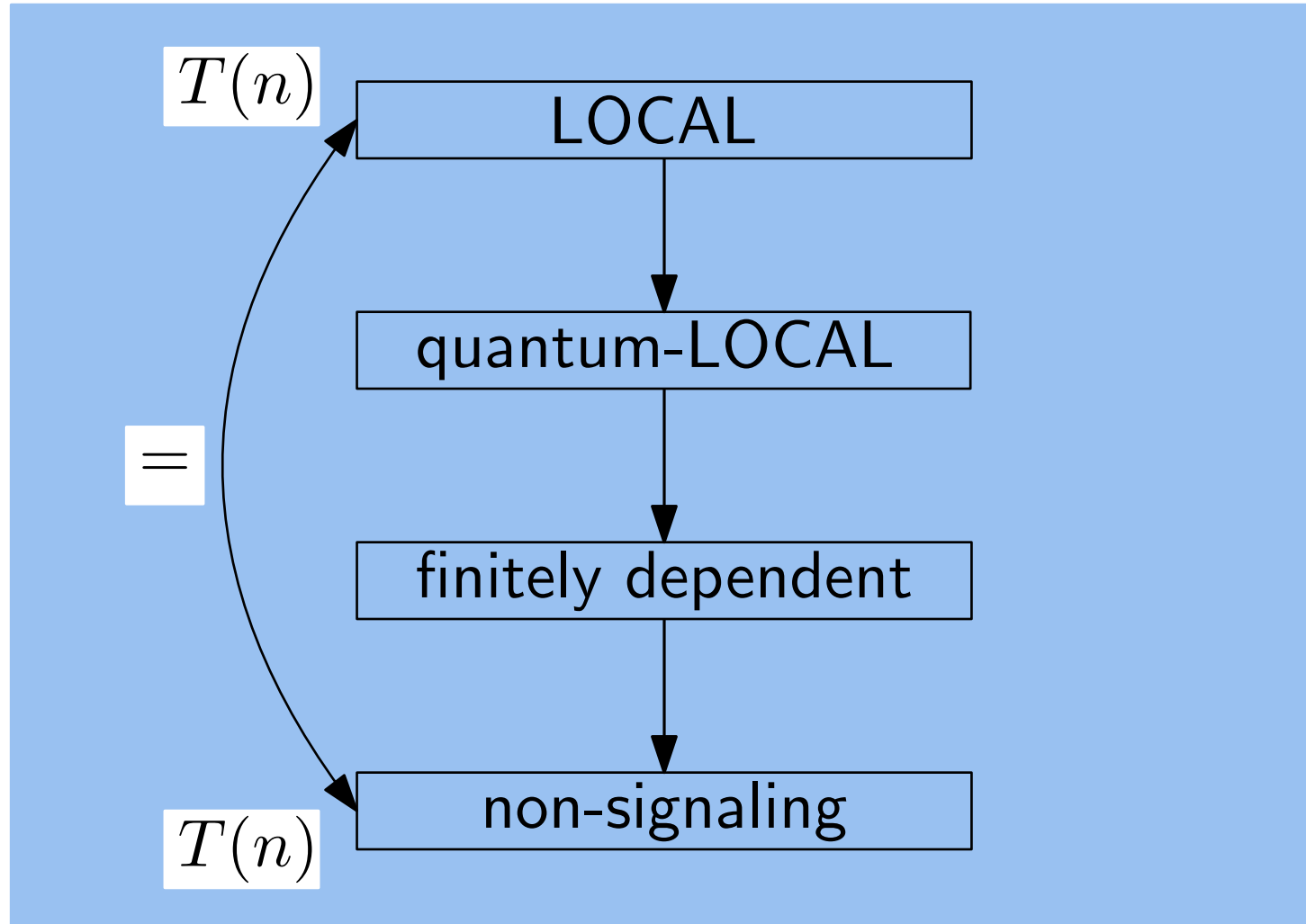
- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E} \left[\sum_{v \in V} X_v \right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α -approx. then $\mathbb{E}[X]$ is an α -approx.

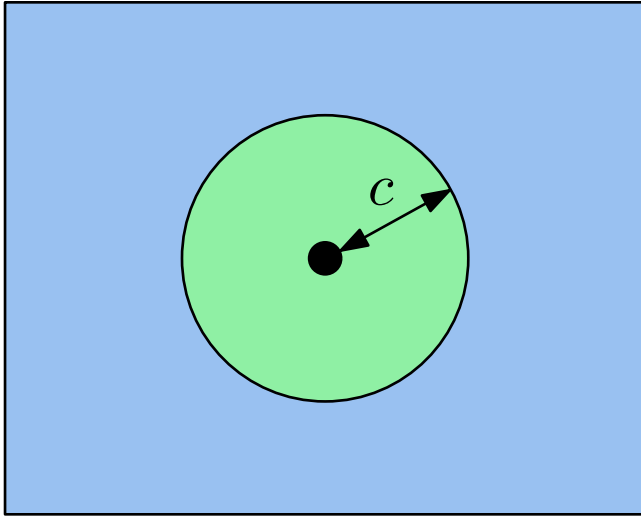
Results for Linear Programs (LPs)

For Linear Programs (LPs)



Results for Locally Checable Labelly Problems (LCLs)

LCLs: explore up to radius c to check a solution



Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Reminder on GHZ

Three players A , B and C , and one referee

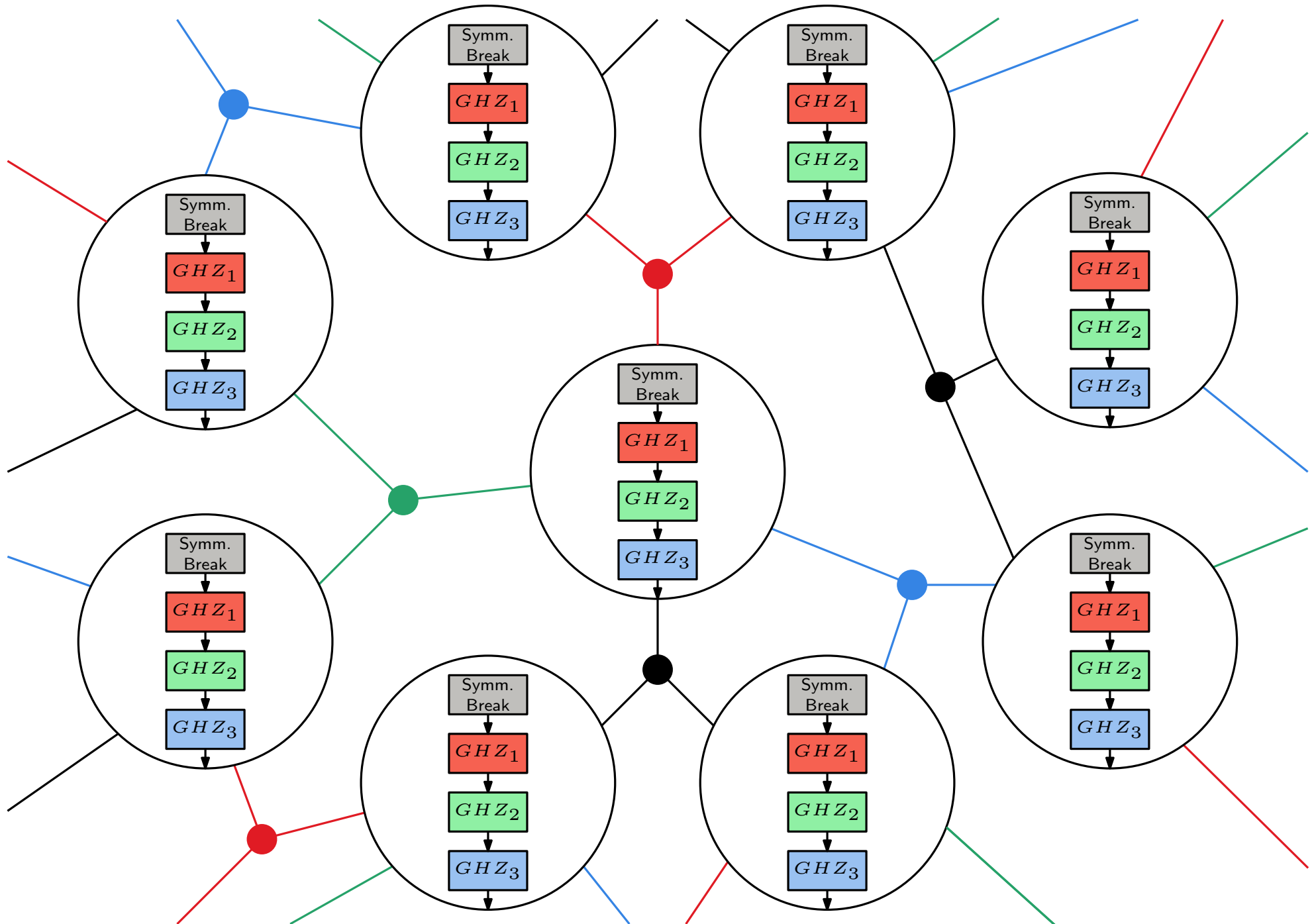
inputs x , y and z ; outputs: a , b and c

Goal: **without communicating after the input is given**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

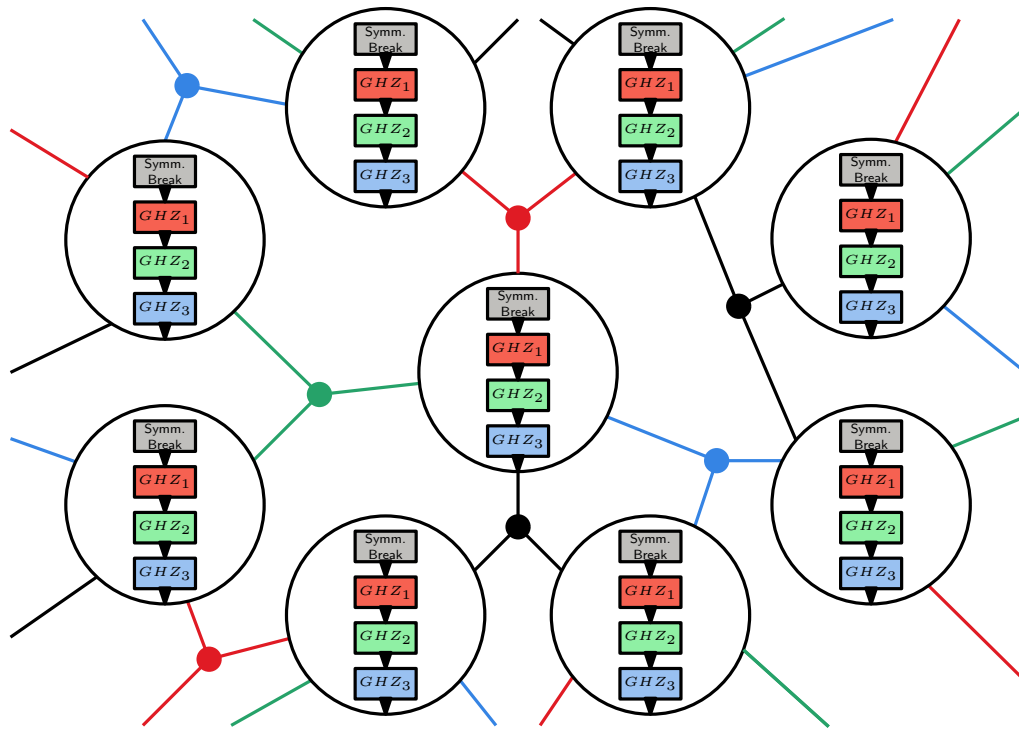
knowing that $x \vee y \vee z \in \{(0, 0, 0), (1, 1, 0), (1, 0, 1), (0, 1, 1)\}$

Results for Locally Checable Labelly Problembs (LCLs)



Results for Locally Checable Labelly Problembs (LCLs)

iterated-GHZ complexity



quantum-LOCAL: $O(1)$ rounds

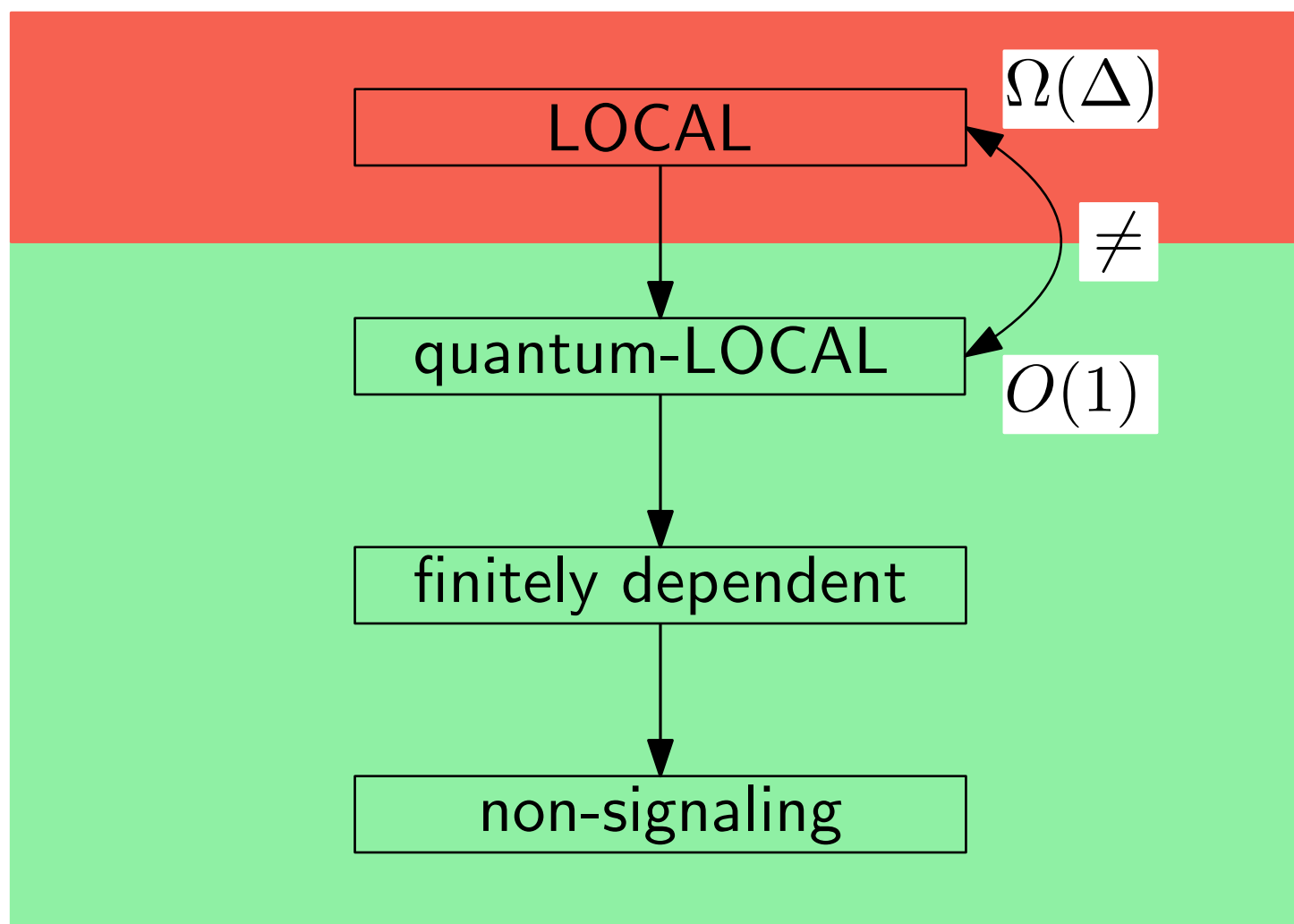
- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1st neigh.
- solve GHZ_1 game with 2nd neigh.
- solve GHZ_2 game with 3rd neigh.
- ...
- solve GHZ_Δ game with the last neigh.

Results for Locally Checkable Labelling Problems (LCLs)

For Locally Checkable Labelling Problems (LCLs)



Summary

Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

- previously assumed to be true but never formalized
- allows us to prove stronger results

LCLs : $O(\Delta)$ LOCAL vs $O(1)$ quantum-LOCAL

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

- only one previous result for LOCAL, and not for LCLs
- most previous literature focused on other models like CONGEST

Summary

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Quantum String Matching

From Bit-Parallelism to Quantum String Matching for Labelled Graphs

M. Equi, H. Lievonen, A. Meijer-van de Griend, V. Mäkinen

CPM 2023

Quantum Pattern Matching in Generalised Degenerate Strings

M. Equi, H. Lievonen, Md R. Islam Khan, V. Mäkinen

arXiv 2026, under submission

Thank you! Questions?